SOFTWARE DEVELOPMENT TRAINING • FINAL CAPSTONE PROJECT

TradeFlow

Intelligent Ethio-Djibouti Trade Corridor Freight Logistics, Telematics & Digital Smart Escrow Platform

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Project Deliverables & Verified Access Links

> **Official Submission Notice:** All project code, server infrastructure, and mobile APK builds are live, fully deployed, and verified. Below are the primary access URLs required for grading:

Deliverable Name	Environment & Platform	Direct URL / Link
GitHub Monorepo	Source Code Repository	https://github.com/beki255/trade-flow
Live Web Application	Vercel Global Edge Network	https://trade-flow-ukgu.vercel.app
Backend REST API (Swagger)	Render Cloud Microservice	https://tradeflow-api-7abs.onrender.com/api/docs
Operational Health Check	Cloud Service Health Monitor	https://tradeflow-api-7abs.onrender.com/api/health
Standalone Android APK	Compiled APK Binary (v1.1.6)	Download TradeFlow v1.1.6 APK
Expo EAS Cloud Build Logs	CI/CD Cloud Build Pipeline	View Expo EAS Build Records

Mobile Android Application QR Code & Download

📲 Instant Smartphone Camera QR Scan

Scan this QR code with any smartphone camera to instantly download and install the standalone **TradeFlow Android App (v1.1.6)**. No Expo Go client or development setup is required. The APK installs natively on Android 10+ devices.

Direct APK Download Link →

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Executive Summary

TradeFlow is an integrated digital freight exchange, real-time corridor telematics, and digital smart escrow platform purpose-built for the critical **Ethio-Djibouti Trade Corridor**. Representing the vital commercial lifeline for landlocked Ethiopia, this 910 km highway and rail artery accommodates over 95% of Ethiopia's international import-export trade volume. Despite its strategic status, corridor operations have historically been severely constrained by informal paper manifests, middleman brokerage commissions charging 15% to 25% markups, chronic payment defaults of 30 to 90 days, a 42% empty return backhaul rate, and severe 24 to 72-hour border gridlocks at Galafi.

TradeFlow systematically overcomes these operational bottlenecks by providing an integrated multi-tenant digital platform:

- **Automated Freight Marketplace:** Connects verified commercial cargo owners directly with fleet operators and truck drivers, eliminating intermediary broker markups.
- **Milestone-Driven Digital Smart Escrow:** Secures 100% of freight payments prior to truck dispatch, automatically disbursing funds to carrier digital wallets upon verified digital proof-of-delivery (e-POD).
- **Corridor Telematics & Geofencing:** Delivers background GPS tracking on mobile driver devices, executing automated waypoint check-ins across 11 strategic corridor transit hubs.
- **Single-Window Electronic Customs Clearance:** Centralizes commercial invoices, customs declarations, packing lists, and cargo transit insurance dossiers with dynamic cryptographic QR transit passes for rapid border clearance.
- **Trilingual Accessibility & Dark Mode:** Ensures complete linguistic inclusion for Ethiopian logistics operators through native English, Amharic (**አማርኛ**), and Afaan Oromoo localization with high-contrast day/night ergonomics.

1. Project Background, Problem Formulation & Scope

1.1 The Ethio-Djibouti Strategic Economic Corridor

With a population exceeding 125 million, the Federal Democratic Republic of Ethiopia is the most populous landlocked nation globally. The nation's foreign trade depends almost entirely upon the maritime gateway provided by the Port of Doraleh in the Republic of Djibouti. The 910 km transit corridor connecting Djibouti to inland container dry port terminals at Modjo and Kaliti handles virtually all incoming industrial capital goods, agricultural fertilizer, energy supplies, and commercial imports, as well as outgoing agricultural commodities (coffee, oilseeds, pulses, and flowers).

1.2 Traditional Freight Logistics Bottlenecks

Field research and logistics industry analysis identify five foundational structural failures across the corridor:

- **Intermediary Brokerage Exploitation:** Freight brokerage is dominated by informal brokers in Addis Ababa and Djibouti who capture 15% to 25% of freight values purely through information asymmetry, inflating consumer prices while leaving transport operators undercompensated.
- **Severe Trust Deficit & Payment Defaults:** Shippers face constant risks of cargo theft, damage, or mid-route freight extortion. Conversely, private truck drivers and small fleet owners face chronic payment defaults or delayed cash disbursements extending 30 to 90 days after delivery.
- **Complete In-Transit Invisibility:** After departing Djibouti, trucks travel for 3 to 7 days through remote desert regions with zero automated tracking. Cargo owners cannot predict arrival times or detect delays, paralyzing industrial manufacturing schedules.
- **Paper-Based Customs Stagnation at Galafi:** Border clearance relies on physical paper manifests and stamps. Drivers wait 24 to 72 hours in extreme desert heat at the Galafi border post, creating immense congestion and truck demurrage costs.
- **The 42% Empty Backhaul Crisis:** Because inbound trucks lack a unified freight exchange to locate outbound agricultural export cargo, over 42% of trucks return to Djibouti completely empty, doubling freight transportation costs.

1.3 Core Project Objectives

The primary engineering objectives achieved by the TradeFlow platform include:

1. **Direct Digital Matching:** Eliminate physical middleman cartels by connecting cargo owners directly with commercial truck fleets.
2. **Payment Solvency via Smart Escrow:** Lock freight payments in digital milestone escrow before truck dispatch, releasing funds automatically upon verified electronic Proof-of-Delivery (e-POD).

3. **Automated Telematics & Waypoint Geofencing:** Provide real-time GPS telemetry, dynamic ETA forecasting, and automated checkpoint check-ins across all 11 critical corridor hubs.
4. **Accelerate Border Customs Clearance:** Implement a single-window digital customs interface with dynamic QR passes to reduce border gate inspection time from days to minutes.
5. **National Linguistic Inclusion:** Deliver complete translations in English, Amharic (**አማርኛ**), and Afaan Oromoo with adaptive dark/light theme ergonomics.

2. Requirements Specification & User Personas

2.1 Five Operational Actor Personas

User Persona	Industry Role	System Permissions & Primary Capabilities
`SHIPPER`	Commercial Importer / Exporter	Publishes freight tenders, funds escrow via Telebirr/CBE, monitors live corridor telemetry, inspects and approves e-PODs.
`TRANSPORTER`	Fleet Owner / Logistics Enterprise	Registers fleet vehicles and drivers, submits competitive freight bids, tracks vehicle earnings, and receives automated escrow payouts.
`DRIVER`	Heavy Commercial Truck Driver	Accesses assigned trips on mobile app, performs GPS geofence check-ins, broadcasts route incidents, captures recipient e-signature & cargo photos.
`CUSTOMS_OFFICER`	Ethiopian Customs Commission (ECC)	Audits electronic dossiers, classifies risk lines (Green/Yellow/Red), issues digital transit passes, scans driver QR codes at gates.
`ADMIN`	TradeFlow System Operator	Verifies business TIN registrations, audits escrow accounts, monitors platform telematics analytics, handles dispute escalations.

2.2 Functional Requirements (FR)

- **FR-01 (Authentication & RBAC):** Multi-tenant authentication with JWT bearer tokens, bcrypt password hashing, and role guards protecting administrative endpoints.
- **FR-02 (Freight Marketplace):** Dynamic tender listings with cargo specification, tonnage, packaging, pickup/delivery locations, and target price in ETB.
- **FR-03 (Carrier Bidding Engine):** Transporters submit bids with equipment specifications and proposed freight rate; shippers award bids with a single click.
- **FR-04 (Digital Smart Escrow):** Mandatory 100% upfront payment locking in milestone escrow prior to dispatch; atomic 95% payout release upon approved e-POD.
- **FR-05 (Corridor Telematics & Geofencing):** Background GPS coordinate collection; automated Haversine geofence check-ins within 15 km of 11 transit hubs.
- **FR-06 (Single-Window Customs):** 4-document electronic dossier upload; algorithmic risk routing (Green Line auto-clear); dynamic QR border pass generation.
- **FR-07 (Electronic Proof of Delivery):** Touchscreen capacitive signature pad and geotagged camera photo capture confirming physical delivery.

- **FR-08 (Crowdsourced Incident Broadcasting):** Drivers report corridor road blockages, accidents, and fuel outages with GPS coordinates and photos.
- **FR-09 (Trilingual Localization):** Instant UI language toggle between English, Amharic (**አማርኛ**), and Afaan Oromoo across web and mobile.
- **FR-10 (Adaptive Ergonomic Theming):** Synchronized Light Mode and High-Contrast AMOLED Dark Mode for desert daylight and nighttime driving.

3. System Architecture & Monorepo Topology

The TradeFlow platform is architected as a modular TypeScript monorepo managed with **pnpm workspaces**. This structure ensures unified type contracts across client and server tiers while keeping concerns cleanly separated:

```
tradeflow-v2/
├── apps/
│   ├── web/              # Next.js 14 App Router, Tailwind CSS, Leaflet GIS, Lucide Icons
│   └── mobile/           # React Native Expo SDK 52, expo-location, SQLite offline queue
├── services/
│   └── api/              # Node.js + Express + TypeScript REST API Microservice
├── packages/
│   └── shared/           # Shared TypeScript interfaces, corridor coordinates, validation
└── docs/                 # Technical documentation, PDF reports, and APK QR assets
```

3.1 Technology Stack Evaluation & Rationale

Layer	Selected Technology	Technical Rationale & Architectural Justification
Web Portal	Next.js 14, React 18, Tailwind CSS	Server-side rendering (SSR), optimized bundle sizes, responsive layout for desktop dispatchers.
Mobile App	React Native, Expo SDK 52	Cross-platform native performance, low battery GPS background tracking, offline camera & canvas signature.
Backend API	Node.js, Express, TypeScript	Non-blocking asynchronous I/O ideal for concurrent GPS telemetry streams, WebSocket broadcasting, and REST endpoints.
Database	PostgreSQL (Neon Serverless)	ACID compliance for escrow financial transactions, type-safe schema migrations, relational integrity.
ORM	Prisma ORM	Declarative schema modeling, automated migrations, type-safe client, and atomic `$transaction` safety.
Cloud Deployments	Vercel (Web), Render (API)	Global Edge CDN distribution for web portals; automated Docker microservice deployment for the Express API.

4. Database Design & Prisma Relational Schema

The persistence layer is modeled using Prisma ORM on PostgreSQL. Normalized schemas prevent data redundancy while enforcing strict foreign key constraints and unique indexing on critical identifiers:

```
// 1. User Account & Role Partitioning (services/api/prisma/schema.prisma)
model User {
  id           String      @id @default(uuid())
  email        String      @unique
  passwordHash String
  fullName     String
  phoneNumber  String      @unique
  role         UserRole    @default(SHIPPER) // SHIPPER, TRANSPORTER, DRIVER, CUSTOMS_OFFICE
  tinNumber    String?     @unique
  isVerified   Boolean     @default(false)
  loads        Load[]      @relation("ShipperLoads")
  bids         Bid[]       @relation("TransporterBids")
  shipments    Shipment[]  @relation("DriverShipments")
  createdAt    DateTime    @default(now())
}

// 2. Freight Tender Listing Model
model Load {
  id              String      @id @default(uuid())
  shipperId       String
  shipper         User        @relation("ShipperLoads", fields: [shipperId], references: [id
  origin          String      // e.g., 'Kaliti Dry Port, Addis Ababa'
  destination     String      // e.g., 'Port of Doraleh, Djibouti'
  cargoType       CargoType   // CONTAINER, BREAK_BULK, DRY_BULK, LIQUID_BULK, REEFER
  weightTons      Float
  targetRateETB   Float
  status          LoadStatus  @default(OPEN) // OPEN, BIDDING, ASSIGNED, IN_TRANSIT, DELIVER
  bids            Bid[]
  shipment        Shipment?
}

// 3. Digital Smart Escrow Financial Ledger
model Payment {
  id             String        @id @default(uuid())
  shipmentId     String        @unique
  shipment       Shipment      @relation(fields: [shipmentId], references: [id])
  totalAmountETB Float
  platformFeeETB Float         // 5% Platform Revenue
  netPayoutETB   Float         // 95% Carrier Disbursement
  paymentMethod  PaymentMethod // TELEBIRR, CBE_BIRR, BANK_TRANSFER
  status         PaymentStatus @default(PENDING) // PENDING, HELD_IN_ESCROW, RELEASED, REFUN
  escrowLockedAt DateTime?
  releasedAt     DateTime?
}
```

5. Digital Smart Escrow Protocol & Settlements

The most severe friction point in Ethiopian freight transport is the mutual trust deficit between cargo owners and transport operators. Shippers hesitate to disburse freight advances to unverified drivers for fear of theft or demurrage. Conversely, carriers refuse dispatch without cash advances to cover fuel and toll expenses, yet frequently face non-payment or delayed settlements of 30 to 90 days after delivery.

TradeFlow resolves this via a programmable digital escrow protocol that locks 100% of the agreed freight payment prior to vehicle dispatch and automatically disburses funds upon cryptographically verified proof of delivery:

```
[DRAFT] ──> [ESCROW_LOCKED] ──> [IN_TRANSIT] ──> [ARRIVED_DESTINATION] ──> [POD_VERIFIE
```

5.1 Atomic Escrow Settlement Implementation

To eliminate financial concurrency errors, double payouts, or orphan records, the escrow release method executes inside an isolated atomic database transaction:

```
// Atomic Escrow Release Implementation (services/api/src/controllers/payment.controller.ts
export async function releaseEscrowPayout(shipmentId: string, paymentId: string) {
  return await prisma.$transaction(async (tx) => {
    const payment = await tx.payment.findUniqueOrThrow({ where: { id: paymentId } });
    if (payment.status !== 'HELD_IN_ESCROW') {
      throw new Error(`Invalid escrow state: ${payment.status}`);
    }

    const platformFee = payment.totalAmountETB * 0.05; // 5% platform commission
    const netPayout = payment.totalAmountETB * 0.95;   // 95% carrier disbursement

    // 1. Transition payment ledger status to RELEASED
    const updatedPayment = await tx.payment.update({
      where: { id: paymentId },
      data: { status: 'RELEASED', releasedAt: new Date(), platformFeeETB: platformFee, netPa
    });

    // 2. Credit transporter digital wallet balance
    await tx.transporterWallet.update({
      where: { userId: payment.shipment.transporterId },
      data: { availableBalanceETB: { increment: netPayout }, lockedEscrowETB: { decrement: p
    });

    return updatedPayment;
  });
}
```

6. Corridor Telematics, GPS Geofencing & Offline Sync

The platform models the authentic road coordinates of the Ethio-Djibouti Corridor (A1 Highway), establishing eleven strategic check-in waypoints between Kaliti Dry Port and the Port of Doraleh:

ID	Waypoint Name	Latitude	Longitude	Km from Origin	Facility Type
`WP-01`	Addis Ababa (Kaliti ICT)	9.005401	38.763611	0 km	Inland Container Depot / Origin
`WP-02`	Modjo Dry Port & Toll	8.591234	39.121543	73 km	Dry Port & Expressway Junction
`WP-03`	Adama Expressway Exit	8.540000	39.270000	99 km	Highway Tollway Interchange
`WP-04`	Metehara Sugar Depot	8.898600	39.919700	190 km	Rest Area & Weighbridge
`WP-05`	Awash Arba Safe Rest	9.143891	40.170562	225 km	Fuel Stop & Truck Parking
`WP-06`	Gewane Desert Hub	10.166400	40.647200	360 km	Emergency Fuel & Repair
`WP-07`	Semera Regional Hub	11.794200	41.008600	588 km	Afar Regional Capital Logistics Hub
`WP-08`	Mille Junction Checkpoint	11.416700	40.750000	515 km	Security & Weighbridge Gate
`WP-09`	Logia Transport Terminal	11.733300	40.966700	575 km	Heavy Truck Fleet Depot
`WP-10`	Galafi Customs Border	11.716700	41.833300	690 km	Ethiopian Customs Commission Gate
`WP-11`	Port of Doraleh (Djibouti)	11.595000	43.060000	910 km	Deep-Water Container Terminal

6.1 Mathematical Haversine Distance & Geofencing Implementation

To verify whether a truck has genuinely arrived at a designated waypoint without relying on manual driver claims, the mobile app executes the spherical **Haversine Formula** on coordinates provided by `expo-location`:

```
 // Corridor Geofence Calculation (apps/mobile/src/shared/corridor.ts)
export function calculateHaversineDistance(
  lat1: number, lon1: number, lat2: number, lon2: number
): number {
  const R = 6371; // Earth's mean radius in kilometers
  const dLat = (lat2 - lat1) * (Math.PI / 180);
  const dLon = (lon2 - lon1) * (Math.PI / 180);
  const a =
    Math.sin(dLat / 2) * Math.sin(dLat / 2) +
    Math.cos(lat1 * (Math.PI / 180)) * Math.cos(lat2 * (Math.PI / 180)) *
    Math.sin(dLon / 2) * Math.sin(dLon / 2);
  const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
  return Math.round(R * c);
}

// Unlocks checkpoint check-in button strictly when within 15 km
export const GEOFENCE_RADIUS_KM = 15;
export function isWithinWaypointGeofence(driverLat: number, driverLng: number, wp: Waypoint)
  return calculateHaversineDistance(driverLat, driverLng, wp.latitude, wp.longitude) <= GEOF
}
```

7. Single-Window Customs Clearance & QR Pass

The Galafi border post separating Ethiopia and Djibouti is the primary bottleneck on the corridor. Historically, drivers carried voluminous physical paper binders containing customs transit declarations, commercial invoices, packing lists, and insurance certificates. Border officers manually cross-referenced serial numbers against manifest logs, resulting in dwell times of 24 to 72 hours per consignment.

TradeFlow integrates a **Single-Window Electronic Customs Dossier** enabling digital upload, validation, and storage of mandatory import-export documentation:

- **Customs Declaration:** PDF/image document confirming official Ethiopian Customs Commission tariff assessment.
- **Commercial Invoice:** Establishes declared CIF/FOB cargo valuation for duty computation.
- **Packing List:** Itemizes container weights, tare, package count, and seal integrity hashes.
- **Transit Insurance:** Guarantees maritime and highway cargo transit liability coverage.

Upon dossier clearance, the mobile app renders a high-contrast cryptographic QR pass on the driver's screen. Border customs officers scan the code using the Customs Officer Web Portal or mobile camera to instantaneously verify consignment validity, truck plate, driver identity, and container seal status in under three seconds.

8. Driver Mobile Application (React Native Expo SDK 52)

The mobile client (`apps/mobile`) is constructed with React Native and Expo SDK 52, specifically engineered for extreme environmental durability, low battery consumption, and offline operation in high-heat desert conditions.

Key driver modules include:

- **Active Trip Dashboard:** Displays consignment details, container numbers, delivery instructions, and completed waypoints.
- **Geofenced Waypoint Check-In:** Proximity-verified check-ins at each of the 11 corridor transit hubs, unlocking only when within 15 km.
- **Crowdsourced Incident Broadcasting:** Enables drivers to report road washouts, customs delays, and fuel outages with geotags and photos.
- **Dual-Factor Electronic Proof of Delivery (e-POD):** Captures recipient touchscreen signature via vector canvas and offloaded cargo photo via camera.
- **Offline Queue Synchronization:** Buffers telemetry pings in local `AsyncStorage` when crossing cellular dead zones, flushing automatically upon network reconnection.

9. Transporter Fleet Management Portal

The Transporter Portal provides fleet operators and logistics companies with a comprehensive operational cockpit across web and mobile:

- **Fleet Registry:** Maintains truck plate numbers, trailer types (Flatbed, Lowbed, Box, Tanker), payload capacities, and driver assignments.
- **Open Loads Marketplace:** Browses open corridor freight tenders published by cargo owners with route and pricing filters.
- **Competitive Bidding:** Submits rate proposals in ETB and committed transit timelines.
- **Escrow Wallet Ledger:** Real-time financial dashboard distinguishing locked escrow from cleared available balances ready for Telebirr/CBE withdrawal.

10. Shipper & Customs Commission Web Portals

Built with Next.js 14 App Router, the web portals deliver high-performance dispatch management:

- **Freight Tender Creation Wizard:** Generates listings with automatic corridor distance calculations and price estimations.
- **Live Leaflet GIS Corridor Map:** Renders active truck markers along the 910 km A1 corridor polyline with live speed and heading indicators.
- **Single-Window Customs Inspection Console:** Pre-arrival manifest queue and digital gate scanner for Ethiopian Customs Commission officers at Galafi.
- **One-Click e-POD Approval:** Instant review of delivery signatures and photos, executing atomic escrow disbursement.

11. Trilingual Localization & Ergonomic Theming

TradeFlow achieves nationwide accessibility through complete native trilingual localization across **English (EN)**, **Amharic (AM - አማርኛ)**, and **Afaan Oromoo (OM)**. Localized strings are stored in structured JSON dictionaries (`locales/en.json` , `locales/am.json` , `locales/om.json`) managed via React Context, enabling instantaneous language toggling without page reload.

Furthermore, an adaptive theme engine synchronizes clean Light Mode with high-contrast AMOLED Dark Mode, resolving contrast issues on devices like the Samsung Galaxy A13 during intense desert sunlight and nighttime driving.

12. Backend REST API Architecture

The backend microservice (`services/api`) is architected using the standard Node.js/Express Controller-Service design pattern, separating HTTP request parsing, business logic execution, and database persistence into decoupled modules.

Key REST endpoints cataloged and documented via interactive Swagger docs (`/api/docs`) include:

- `POST /api/auth/register` & `POST /api/auth/login`: Authentication and JWT token issuance.
- `GET /api/loads` & `POST /api/loads`: Freight marketplace browsing and tender creation.
- `POST /api/bids` & `POST /api/bids/:id/accept`: Carrier bidding and bid awarding.
- `POST /api/payments/escrow/lock` & `POST /api/payments/escrow/release`: Milestone escrow locking and release.
- `POST /api/telematics/ping` & `POST /api/shipments/:id/checkin`: Mobile GPS coordinate streaming and waypoint check-in.
- `POST /api/customs/dossier` & `GET /api/customs/qr/:shipmentId`: Electronic customs dossier upload and QR transit pass generation.
- `GET /api/health`: Real-time service uptime, memory metrics, and PostgreSQL connection status.

13. Testing, Verification & Quality Assurance

TradeFlow underwent multi-tiered verification covering automated unit testing, end-to-end integration workflows, escrow financial concurrency tests, and physical hardware deployment across real Android smartphones.

Test ID	Subsystem	Test Procedure & Test Data	Expected Behavior	Status
TC-01	Auth & RBAC	Register 5 users across each role	JWT issued; role guards restrict unauthorized routes.	PASS
TC-02	Marketplace	Shipper publishes 35-ton coffee load	Tender appears on transporter console within 200 ms.	PASS
TC-03	Bidding	Transporter submits 145,000 ETB bid	Bid persists in PostgreSQL and alerts shipper.	PASS
TC-04	Smart Escrow	Deposit 145,000 ETB via Telebirr mock	Payment status transitions to `HELD_IN_ESCROW`.	PASS
TC-05	Geofencing	Simulate GPS at Semera (11.7942, 41.0086)	Check-in button enables within 15 km threshold.	PASS
TC-06	Geofence Reject	Simulate GPS 45 km away from waypoint	Check-in button disabled; displays distance warning.	PASS
TC-07	Customs QR	Generate Green Line transit pass	High-contrast QR renders with valid cryptographic hash.	PASS
TC-08	Gate Scanner	Scan QR with Customs Officer portal	Displays cargo manifest and marks pass CLEARED.	PASS
TC-09	e-POD Signature	Capture signature on touch canvas	Exports base64 PNG vector and timestamps record.	PASS
TC-10	Atomic Payout	Shipper approves completed e-POD	Atomic release: 95% (137,750 ETB) to carrier; 5% fee retained.	PASS
TC-11	Localization	Toggle language EN -> AM -> OM	100% of labels translate instantly with zero reload.	PASS
TC-12	Offline Queue	Cut mobile network, log 5 GPS pings	Pings queue in AsyncStorage; auto-sync when online.	PASS

14. Technical Challenges Encountered & Applied Solutions

1. **Desert Cellular Intermittency & Telemetry Data Loss:** When trucks travel through remote areas in the Afar desert between Awash, Gewane, and Semera, cellular signals frequently drop for up to 120 km. To solve this, an `AsyncStorage` local FIFO queue caches telemetry pings with GPS timestamps, flushing automatically via exponential backoff upon network restoration.
2. **Samsung Galaxy A13 AMOLED Dark Mode Text Inversion:** Samsung One UI forced background color overrides on static styles, producing unreadable black text on dark cards. This was resolved by re-architecting all mobile styles into dynamic React factory hooks (`getDriverStyles(theme)`) wrapped inside `useMemo` .
3. **Escrow Concurrency Race Conditions & Double Disbursements:** Rapid double-clicking on the "Approve e-POD" button could trigger concurrent payout requests. This was resolved by executing payout updates within atomic `prisma.$transaction` blocks with strict state pre-checks.
4. **React Native Expo SDK 52 Standalone APK Native Compilation:** Cloud builds initially encountered native linking errors with signature canvas and camera modules. This was resolved by configuring custom Expo config plugins and memory allocations in `eas.json` .
5. **Haversine Distance False Positives in Mountainous Terrains:** Straight-line Euclidean proximity triggered premature check-ins on winding mountain passes near Metehara. This was mitigated by combining the 15 km threshold with strict chronological waypoint sequencing.

15. Conclusion & Future Outlook

The **TradeFlow** platform establishes an authentic, production-grade foundation for cross-border freight logistics across the Ethio-Djibouti Trade Corridor. By replacing manual brokerages with an automated freight exchange, securing carrier freight payments via milestone-locked digital escrow, delivering continuous GPS telematics, and accelerating customs clearance through single-window QR passes, TradeFlow bridges critical operational gaps on Ethiopia’s foremost trade lifeline.

All software deliverables have been constructed, tested, and verified against the academic requirements of the Software Development Training Capstone. Future iterations will incorporate direct OBD-II hardware telematics dongles for fuel monitoring and automated plate recognition (ANPR) cameras at Galafi border checkpoints.